Who builds what, and by when

Three owners, one plan to the 31 December release. Dates follow the System Development Plan. Tasks marked **late** were due before 21 October. The live tracker is the up-to-date version; this PDF is a snapshot.

Elisha Lungu
API & infrastructure

Django, database, sign-in, sync, advisory content, servers, CI/CD, security.

`backend/ .github/`

17 tasks · 5 done

Agness
ML & model delivery

Images, labelling, training, conversion, and the in-app inference module.

`ml/ pwa/src/inference/`

16 tasks · 0 done

Robert Ng'onga Chishangila
PWA & field UX

Screens, camera, offline storage and sync queue, languages, voice, field testing.

`rest of pwa/`

15 tasks · 0 done

Milestones

ID	Milestone	Due	Owner(s)	State
M1	Contract, designs, class list signed off	14 Oct	Agness, Robert	**Late**
M2	6,000+ field images	28 Oct	Agness	Open
M3	Dataset v1 + baseline model	11 Nov	Agness	Open
M4	≥85% model under 15 MB	25 Nov	Agness	Open
M5	Full offline PWA	9 Dec	All three, Robert	Open
M6	Field validation	23 Dec	All three	Open
M7	Public MVP release	31 Dec	All three, Elisha, Robert	Open

Where your work meets someone else's

- **Elisha ↔ Robert:** `docs/api-contract.yaml` and `docs/offline-sync.md`.
- **Agness ↔ Robert:** the function `diagnose(image) → Prediction`, shape in `docs/model-output.md`. Robert builds against a fake version until the real model lands.
- **Agness ↔ Elisha:** the model manifest, `labels.json`, and where model files are hosted.
- Change any of these only in a pull request the other side approves.

Working rules

- Every pull request gets one approval from the owner of the code it touches; `docs/` needs all three.
- Branches live 2–3 days at most. Merge small and often.
- Friday demo: each person shows one thing working on a real phone or real data.
- If Robert is behind by Week 7, Elisha takes the service worker and sync queue.

ML & MODEL DELIVERY

Agness

16 tasks, 0 done. Sorted by due date.

Due	Task	Step	Status
23 Oct	**Write the one-page photography protocol** Distance, angle, light, background, resolution, colour card, and the metadata CSV columns in ml/README.md.	Step 3	☐ To do
24 Oct	**Agronomist sign-off on the 19-class disease list** docs/class-taxonomy.md, five questions at the bottom. Labelling can't start without it.	Step 2 **M1**	☐ To do
24 Oct	**Agree diagnose(image) → Prediction with Robert and ship a fake** pwa/src/inference/: a stub returning realistic Predictions so the screens can be built now.	Step 10	☐ To do
28 Oct	**Field collection: 6,000+ images with metadata** Three agro-ecological zones, early/mid/advanced stages plus healthy. Back up raw photos the same evening. All three attend at least one trip.	Step 3 **M2**	☐ To do
28 Oct	**Gather open datasets as a stopgap and map them to the class list** So the baseline isn't blocked if field collection runs short.	Step 3	☐ To do
31 Oct	**TF.js conversion check on an untrained model** EfficientNetB0, float16 export, check size, load in the browser, outputs match Python.	Step 7	☐ To do
1 Nov	**Label Studio with two-expert review** Third expert resolves disagreements.	Step 4	☐ To do
4 Nov	**Dataset manifest and split by farm and field** Never a random shuffle. Versioned with DVC, with unit tests.	Step 4	☐ To do
4 Nov	**Preprocessing and augmentation pipeline** tf.data; harsh sun, hand shake, off-angle shots.	Step 5	☐ To do
11 Nov	**Baseline model and per-class report** Frozen EfficientNetB0 + new head; precision, recall and F1 per class.	Step 6 **M3**	☐ To do
18 Nov	**Fine-tuning, class weighting and confidence calibration**	Step 6	☐ To do
18 Nov	**Second collection round for weak classes** Driven by the confusion matrix from the baseline.	Step 3	☐ To do
25 Nov	**Final model: float16 TF.js under 15 MB** ≥85% held-out accuracy, no class under 70% recall, published with labels.json.	Step 7 **M4**	☐ To do
2 Dec	**Real inference in a Web Worker** Model cached in IndexedDB, download progress with resume, shutter-to-result ≤5 s on a 1 GB phone.	Step 10	☐ To do
2 Dec	**Low-confidence path and tuned thresholds in the manifest**	Step 10	☐ To do
16 Dec	**Field accuracy tool: app diagnosis vs expert diagnosis**	Step 13	☐ To do

API & INFRASTRUCTURE

Elisha Lungu

17 tasks, 5 done. Sorted by due date.

Due	Task	Step	Status
15 Oct	**Data models: farmers, farms, diseases, advice, diagnoses**	Step 8	☑ Done
15 Oct	**Phone + SMS code sign-in with rotating tokens**	Step 8	☑ Done
15 Oct	**Profile and farms endpoints**	Step 8	☑ Done
15 Oct	**Diagnosis sync endpoint, idempotent on client_id**	Step 8	☑ Done
15 Oct	**Advisory bundle, model manifest, feedback and history endpoints built** On branches, waiting for review (see Week 4).	Step 8	☑ Done
23 Oct	**Add CODEOWNERS and branch protection** Owner-approval per folder; docs/ needs all three. Protect main and develop.	Step 1	☐ To do
24 Oct	**Review and merge the five open branches** Advisory bundle, model manifest, feedback, history, offline-sync doc. Agness or Robert reviews.	Step 8	☐ To do
28 Oct	**Concurrency test for the sync race on PostgreSQL** Two requests with the same client_id at once; covers the IntegrityError path in sync.py.	Step 8	☐ To do
4 Nov	**Staging server** Docker, PostgreSQL/PostGIS, HTTPS.	Step 14	☐ To do
4 Nov	**Connect sign-in to a live SMS provider on staging**	Step 8	☐ To do
11 Nov	**Advisory text for all 19 classes, sent for agronomist review** Treatment, dosage, safety warning, prevention, follow-up.	Step 8	☐ To do
18 Nov	**Sentry, uptime monitoring and rate limits on staging**	Step 13	☐ To do
25 Nov	**Model hosting on a CDN and a command to publish a release**	Step 7	☐ To do
2 Dec	**Approved advisory bundle live on staging**	Step 8	☐ To do
9 Dec	**CI deploys develop to staging; API docs and Postman collection**	Step 14	☐ To do
16 Dec	**Security review** Rate limits, input validation, dependency audit, secrets rotation, backups.	Step 13	☐ To do
30 Dec	**Production servers, backups, TLS and monitoring**	Step 14 **M7**	☐ To do

PWA & FIELD UX

Robert Ng'onga Chishangila

15 tasks, 0 done. Sorted by due date.

Due	Task	Step	Status
24 Oct	**Figma: the seven screens** Capture, processing, result, advice, history, settings, language.	Step 2 **M1**	☐ To do
28 Oct	**App shell, routing and mobile layout**	Step 9	☐ To do
4 Nov	**Camera capture** Live preview, framing guide, gallery-upload fallback.	Step 9	☐ To do
4 Nov	**Image checks and resize** Blur, brightness, minimum resolution; centre-crop to 224×224.	Step 9	☐ To do
11 Nov	**Sign-in screens against staging**	Step 9	☐ To do
11 Nov	**Result and advice screens** Built on the fake diagnose() and the real advisory bundle.	Step 9	☐ To do
18 Nov	**History, settings and install prompt**	Step 9	☐ To do
25 Nov	**Test the PWA on a 1 GB Android phone and fix slow screens**	Step 9	☐ To do
2 Dec	**Service worker precaching app shell, model and advice**	Step 11	☐ To do
9 Dec	**Offline history and sync queue** IndexedDB, retry with backoff, offline badge and pending-sync count. Elisha pairs on this.	Step 11 **M5**	☐ To do
9 Dec	**Translations and language selector** i18next with en, ny, bem, toi files; terms reviewed by an extension officer.	Step 12	☐ To do
16 Dec	**Voice narration and recorded audio in four languages**	Step 12	☐ To do
16 Dec	**Accessibility pass and Lighthouse PWA score ≥90**	Step 13	☐ To do
16 Dec	**Field test scripts and observation sheets**	Step 13	☐ To do
31 Dec	**Production PWA release, checked on several phones**	Step 14 **M7**	☐ To do

All three

6 tasks, 1 done. Sorted by due date.

Due	Task	Step	Status
13 Oct	**Repo foundations and first API contract** Monorepo, CI, README, contributing rules, OpenAPI draft, class list draft.	Steps 1–2	☑ Done
23 Oct	**Decide: photo uploads, and phone-number login** The deck promises no uploads and no typing; the API has an optional photo upload and a phone + code login. Pick one answer for each and update the contract or the deck.	Step 2	☐ To do
9 Dec	**Aeroplane-mode test from a cold start** Install, diagnose, store, then sync on reconnect.	Step 11 **M5**	☐ To do
23 Dec	**Field validation in two districts** 30–50 pilot farmers and 5 extension officers; record field accuracy and every point of confusion.	Step 13 **M6**	☐ To do
23 Dec	**Fix every critical and high defect from the field**	Step 13 **M6**	☐ To do
31 Dec	**User guide, extension officer pack and iHub report**	Step 14 **M7**	☐ To do